

Deliverable 3 — Use-Case Flow Specification

UC-01 Customize Monthly Box • Customizable Subscription Box Scheduler

Name	SRN	Course
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Primary Actor	Subscriber
Secondary Actor	Catalog / Inventory Service (supporting)
Scope & Level	Customizable Subscription Box Scheduler (portal) - user-goal level
Trigger	Subscriber opens "My Next Box" and selects Customize.
Traces to	FR-001, FR-002, NFR-002
Relationships	«include» UC-06 Validate Customization Window; extended by «extend» UC-09 Suggest Substitute Item

Preconditions

1. The Subscriber is authenticated and the subscription status is Active (not Suspended or Cancelled).
2. A box exists for the current cycle and has not yet been locked for billing by UC-04.
3. Server time is more than 48 hours before the subscription's renewal timestamp (UTC).
4. The catalog for the cycle is published and live stock levels are available.

Postconditions

1. The box item list is persisted with the new selections and an incremented box version number.
2. An audit record (subscriber, old contents, new contents, timestamp) is written.
3. The cycle's fulfillment manifest is flagged for refresh within 5 minutes.
4. A confirmation showing the new contents and the remaining customization window is sent to the Subscriber.
5. Failure guarantee: on any error the previously confirmed contents remain intact, stock reservations are released, and no manifest refresh is triggered.

Main Success Scenario

1. Subscriber selects "Customize" on the next scheduled box.
2. System performs «include» UC-06 Validate Customization Window: it computes the time remaining to the renewal timestamp and confirms it is greater than 48 hours.
3. System displays the current box contents, a countdown to the customization cut-off, and the catalog filtered by the Subscriber's preference tags (FR-002).
4. Subscriber removes one item from the box and selects a replacement SKU from the filtered catalog.
5. System checks live stock for the chosen SKU and reserves one unit against the cycle.
6. Subscriber repeats steps 4-5 until the box holds the item count contracted by the plan, then selects "Save changes".
7. System validates the box against the plan: item count, exclusion tags (e.g. nut-free), and price ceiling.
8. System persists the new contents, increments the box version, and writes the audit record.
9. System flags the cycle's fulfillment manifest for refresh and displays a confirmation with the new contents and the customization deadline.
10. Use case ends successfully.

Alternate Flows

5a. Selected item is out of stock (extension point: item unavailable -> «extend» UC-09 Suggest Substitute Item)

- 5a1. The stock check at step 5 returns zero available units for the chosen SKU.
- 5a2. System marks the SKU "Out of stock" and offers up to 3 substitutes carrying the same preference tags.
- 5a3. Subscriber selects a substitute; the flow resumes at step 5 with the substitute SKU.
- 5a4. If the Subscriber rejects every substitute, the original item is left in the box, the system shows "No change made to this slot", and the flow resumes at step 6.

2a. Customization window has already closed (less than 48 h to renewal)

- 2a1. UC-06 Validate Customization Window returns status CLOSED.
- 2a2. System displays the box read-only with the message "This box locked at [renewal timestamp]; changes now apply to your next box."
- 2a3. System offers two onward options: "Customize next month's box" and "Skip next cycle" (UC-03).
- 2a4. If the Subscriber picks either option the corresponding use case starts; otherwise the use case ends with no change to the box.

Exception Flow

- 8a. Persistence failure - the save transaction is rolled back, all stock reservations from step 5 are released, an error with a Retry action is shown, and the box remains at its previous version.